Practice Exam: Genetics (Version 3)

Name: _____

Question type	Points	Time budget
24 multiple choice	3 each (72)	32 min
6 short answer	2 each (12)	6 min
1 essay	25	12 min

Closed book; no calculator needed. The answer key is a separate file.

Multiple Choice

1. At the end of **meiosis I**, each cell has:

A. 1N chromosomes, 2X DNA
B. 2N chromosomes, 4X DNA
C. 1N chromosomes, 1X DNA
D. 2N chromosomes, 2X DNA

2. A cell contains **6 single-rod chromosomes**: 2 large, 2 medium, 2 small. In each size, one came from the mother and one from the father. The cell is:

A. Haploid, replicated
B. Haploid, unreplicated
C. Diploid, replicated
D. Diploid, unreplicated

3. Which is true of **both** mitosis and meiosis II?

A. Homologous chromosomes separate
B. Crossing over occurs
C. Sister chromatids separate
D. Chromosome number is halved
E. Tetrads form

4. Which event is the **physical basis** of Mendel's law of independent assortment?

A. DNA replication in S phase
B. Crossing over in prophase I
C. Sister chromatids separating in anaphase II
D. Homologs separating in anaphase I
E. Random alignment of homologous pairs in metaphase I

5. Two **pink** four-o'clock flowers (Rr, incomplete dominance) are crossed. The phenotypic ratio of offspring is:

A. 3 red : 1 white
B. 1 red : 2 pink : 1 white
C. 1 pink : 1 white
D. 9 : 3 : 3 : 1

6. Which is a possible gamete from an **AaBbDd** individual (unlinked genes)?

A. AbD
B. AaB
C. ABb
D. AaBbDd
E. Ab

7. A woman who is a **carrier** for hemophilia (X-linked recessive) has children with an unaffected man. What is the probability that a **son** is affected?

A. 0
B. 1/4
C. 1/2
D. 1

8. In a pedigree, **two affected parents** have an **unaffected daughter**. The trait is most likely:

A. Autosomal recessive
B. X-linked recessive
C. Not genetic
D. Autosomal dominant

9. Genetic linkage is an exception to which of Mendel's laws?

A. Segregation
B. Independent assortment
C. Dominance
D. Particulate inheritance

10. A dihybrid F2 cross gives **9 black : 3 brown : 4 white**. This most likely indicates:

A. One gene with incomplete dominance
B. Many genes adding up (polygenic)
C. Co-dominance
D. Complete linkage with no crossing over
E. Gene-by-gene interaction (epistasis)

11. A dominant allele makes a working enzyme; the recessive allele makes a non-working version. In a **heterozygote**:

A. Both alleles are typically transcribed, but the working enzyme determines the phenotype
B. Only the dominant allele is transcribed
C. The recessive allele is removed during meiosis
D. The phenotype is always intermediate

12. Which is an example of **multiple allelism**?

A. Human height
B. PKU affecting many unrelated traits
C. ABO blood type
D. Coat color with a 9:3:4 ratio

13. A DNA sample is **30% adenine**. What percent is **cytosine**?

A. 30%
B. 70%
C. 40%
D. 20%
E. 60%

14. DNA starts fully heavy (^{15}N) and replicates **semi-conservatively** for **two rounds** in ^{14}N. What fraction of DNA molecules are **intermediate** density?

A. All of them
B. One half
C. One quarter
D. None

15. Which strand is made as **Okazaki fragments**, and why?

A. The leading strand, because polymerase builds 3' to 5'
B. The leading strand, because it has no primer
C. The lagging strand, because polymerase only builds 5' to 3' and that template runs opposite to fork movement
D. The lagging strand, because helicase skips it

16. Transcription is the process of:

A. DNA to RNA
B. RNA to protein
C. DNA to DNA
D. RNA to DNA
E. Protein to RNA

17. A mutation in the lac **operator** prevents the repressor from binding. Glucose is absent. The lac genes will be:

A. Never transcribed
B. Transcribed only when glucose is present
C. Transcribed only when lactose is present
D. Transcribed only at very low levels
E. Transcribed at high levels even without lactose

18. The **template strand** reads **3'-TACGGT-5'**. The mRNA is:

A. 3'-AUGCCA-5'
B. 5'-TACGGT-3'
C. 5'-UACGGU-3'
D. 5'-AUGCCA-3'

19. Which describes a typical **prokaryotic** genome?

A. Many linear chromosomes containing introns
B. One circular chromosome, mostly genes, with operons
C. Mostly transposable elements
D. Size unrelated to metabolic ability

20. About what fraction of the human genome is **protein-coding exons**?

A. About 50%
B. About 26%
C. Less than 2%
D. About 80%
E. About 20%

21. What are the first three amino acids translated from **5'-CCAUGGCUUGGUAG-3'**? (Use the codon table.)

A. Met-Ala-Trp
B. Pro-Trp-Leu
C. Met-Gly-Leu
D. His-Gly-Leu

22. Which mutation would most likely cause the **MOST severe** change to a protein?

A. A third-position change that doesn't change the amino acid
B. A missense change from Leu to Ile (both hydrophobic)
C. A 3-nucleotide deletion near the end of the gene
D. A single-nucleotide insertion in codon 2

23. Which happens in **prokaryotes but NOT eukaryotes**?

A. RNA splicing
B. Coupled transcription and translation
C. Adding a 5' cap
D. Adding a poly(A) tail
E. Alternative splicing

24. A patient makes a **normal** protein but in **very low amounts**. Exome sequencing finds no mutation. The mutation is most likely in:

A. An exon, creating a stop codon
B. An exon, causing a frameshift
C. The promoter
D. The start codon

Short Answer

Define each term in a **single sentence** (2 pts each).

25. Homozygous
26. Carrier
27. Promoter
28. Okazaki fragments
29. Nondisjunction
30. Pleiotropy

Essay (25 pts)

E. coli can use lactose as food, but only makes the lac operon proteins when they are needed.

1. Name the three structural genes of the lac operon and what each protein does. Explain why bacteria organize these genes into an **operon**.

2. Describe how the **lac repressor** and **CAP-cAMP** control transcription. Predict the level of transcription (off, low, or high) for all four combinations of glucose (+/-) and lactose (+/-), and explain each.

3. Predict what happens to lac operon expression if (a) **lacI** is deleted, (b) the repressor **cannot bind lactose**, and (c) **CAP** is non-functional. Explain each.

4. Eukaryotes don't have operons. Explain how a eukaryotic cell turns on many related genes at once (give an example), and name **two** layers of gene regulation that eukaryotes have but prokaryotes lack.

Essay (continued)

Codon Table

1st \ 2nd	U	C	A	G
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA STOP UAG STOP	UGU Cys UGC Cys UGA STOP UGG Trp
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg
A	AUU Ile AUC Ile AUA Ile AUG Met (Start)	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly